PaddyGuard AI:

A Multimodal AI Based System for Rice Leaf Disease And Pest Detection, Treatment Recommendation And Knowledge Assistance



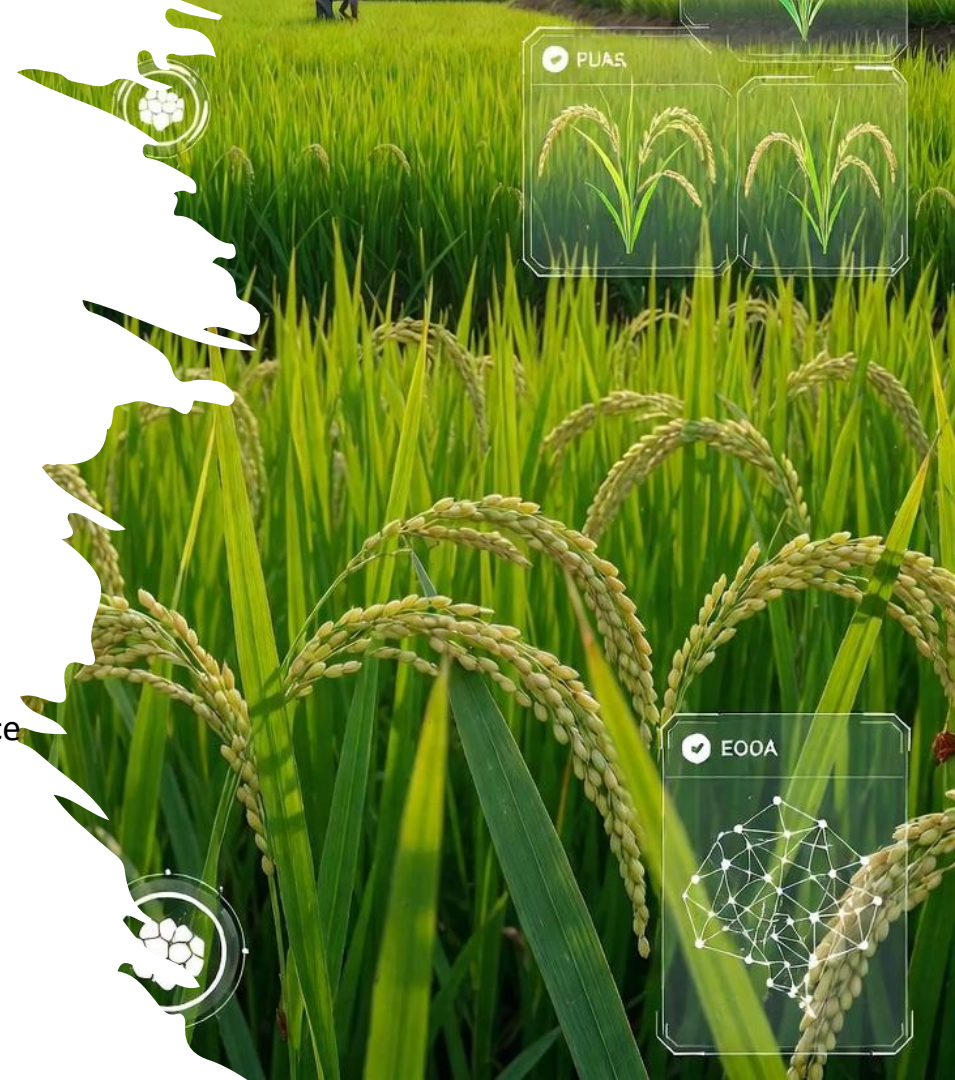
Project ID : R26-SE-015

Introduction

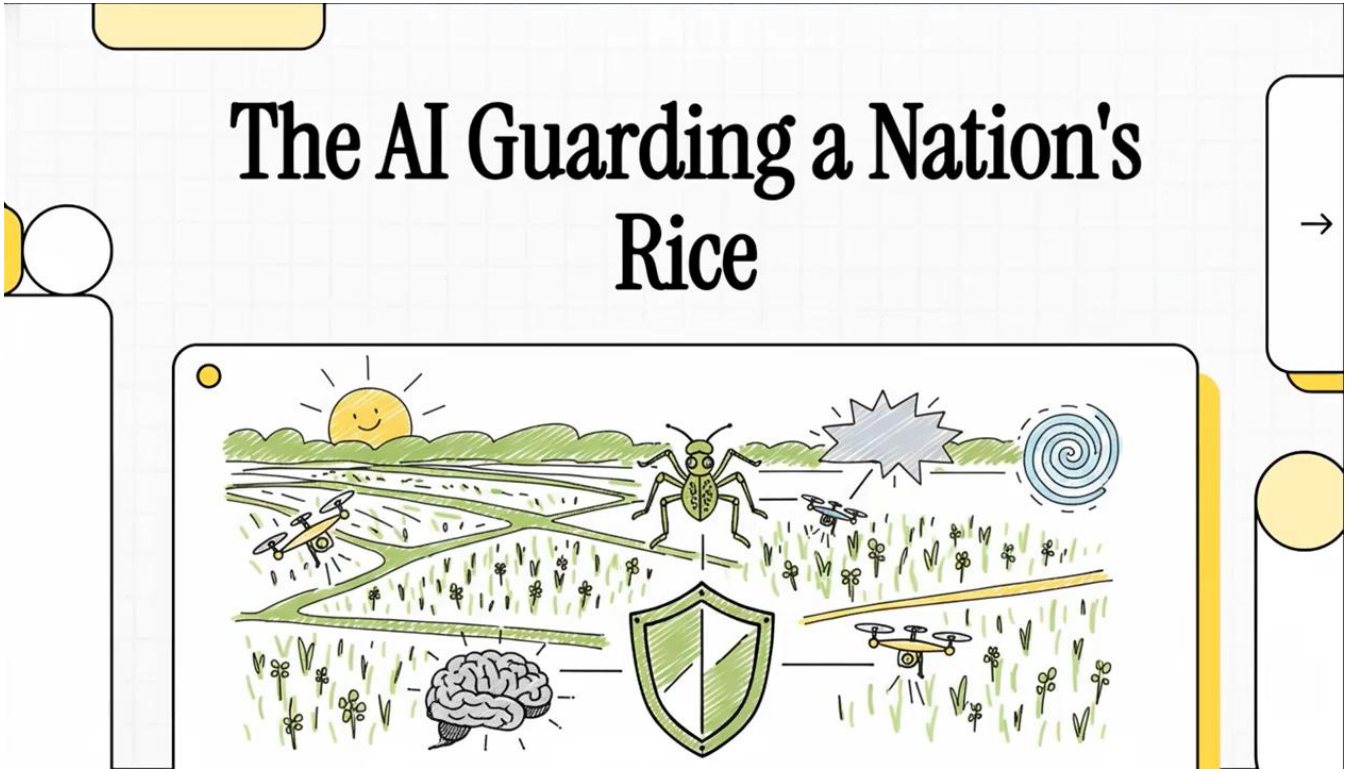
- Rice is the backbone of Sri Lanka: feeds 21M people; 1.8M households rely on paddy farming.
- Challenges: diseases like Bacterial Blight, Rice Blast, Brown Spot can destroy 30% - 80% of harvest; limited access to experts.
- Existing tools limitations: mostly image-based, English-only, not farmer-friendly, require good cameras.

PaddyGuard AI solution:

- PaddyGuard AI addresses all of these gaps through voice and image-based diagnosis, Sinhala-language support, and treatment advisory - directly aligned with SDG 2, Zero Hunger, and SDG 9, Innovation and Infrastructure.
- Sinhala-compatible advisory chatbot.
- Easy-to-use, mobile-friendly platform.



Problem Definition



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Comments for the Proposal and How the Team Addressed Them

Comment Received	How We Addressed It	Status
Expand dataset with more complex and regional dialect data	Created 5000+ row Sinhala symptom dataset covering 5 regional accents (Southern, Uva, North Central, Sabaragamuwa, Central) with expert-validated agricultural vocabulary.	Resolved
Strengthen model comparison and baseline justification	Conducted 6-model comparative study (NB, LR, RF, SVM, DistilBERT, BERT). SVM selected as best with 96.67% F1 — justified with size and speed trade-off analysis.	Resolved
There are no novelties for the leaf disease detection component.	PaddyGuard provides two fine-tuning options: offline Google Colab training for controlled research, and in-app PyTorch fine-tuning for automated model improvement.	Resolved
There are no novelties for the pest identification component.	The proposed system can adapt to a newly identified rice pest using only 5–20 labelled images, without retraining the entire model from scratch.	Resolved
Use software engineering principles when developing the project	Adopted formal SDLC practices including requirements specification, UML diagrams, modular architecture, version control (Git), code reviews, and unit/integration testing throughout development.	Resolved
Integrate voice component with treatment advisory system	FastAPI /diagnose endpoint returns disease + confidence to Component 3 (Treatment) via shared backend API contract. Integration tested end-to-end.	Resolved

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Problem & Impact / Gravity

30 – 80%	crop loss from rice diseases when untreated
1.8M	paddy farming households in Sri Lanka affected
21M	people fed by Sri Lankan rice production
0	existing tools supporting Sinhala voice input

Who Benefits

Smallholder paddy farmers · Agricultural extension officers · Department of Agriculture Sri Lanka · Agri-input companies · Crop insurance providers

Design Excellence of the Overall Solution

C1 — Voice NLP

Sinhala ASR + 6-model comparison. SVM 96.67% F1. First agricultural Sinhala voice-NLP pipeline with 3-signal OOD detection.

C2 — Leaf Image

CNN multi-architecture comparison (ResNet50, EfficientNetB3, DenseNet121). 92%+ accuracy with Grad-CAM explainability + ODIN OOD.

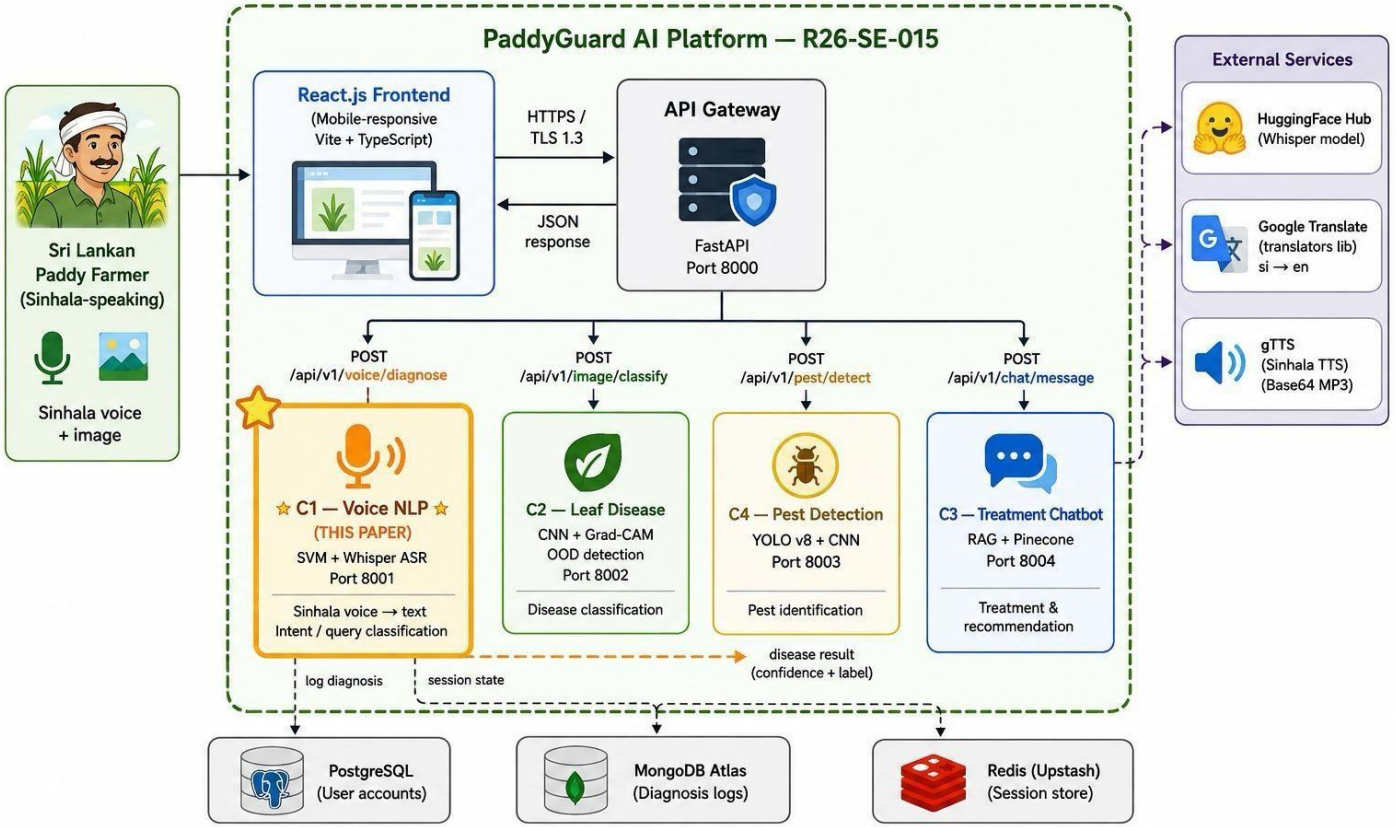
C3 — Treatment

RAG-based chatbot (IRRI + FAO knowledge) with rule-based advisory engine. Sinhala language treatment delivery.

C4 — Pest Detect

YOLO v8 + CNN (DenseNet121 best) for 5 pest classes. OOD detection for unknown insects.

PaddyGuard AI: High-Level Architecture Diagram.



Market, Value Proposition And Commercialization

Value Proposition

- Farmers: Instant Sinhala voice-based disease diagnosis & treatment advice.
- Government: Scalable digital advisory for rural farmers.
- Agri Companies: Targeted product recommendations at diagnosis stage.
- Society: Reduced crop losses and improved food security.

Target Market

- Primary: Department of Agriculture & extension services.
- Secondary: Agricultural cooperatives, crop insurance, agri-input companies.
- End Users: ~1.8M paddy farming households in Sri Lanka.

Commercialization

- Government licensing as national agricultural advisory tool.
- API integration for agri-input companies.
- Grants & SaaS analytics for cooperatives and regional expansion.



Thank You